Jiachen Yao Ph.D. Student

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Education

Ph.D. in Computing and Mathematical Sciences — Pasadena, California
California Institute of Technology — 2024 - 2029 *(expected)*

- Advisor: Professor Animashree Anandkumar
- Research area: Generative models, Dynamical Systems, and Operator Learning.

B.S. (Honours) in Information and Computer Science — Beijing, China
Zhili College, Tsinghua University — 2020 - 2024

- Advisor: Professor Jun Zhu
- Outstanding Graduates of Beijing

Research Highlights

Generative sampling and inverse problems

- FunDPS: function-space diffusion posterior sampling with multi-resolution sampling; from 3% observed points, 32% lower error than prior methods with 4× fewer steps.
- DDIS: theory showing why diffusion guidance fails under sparse observations, and a decoupled sampler for inverse problems that fixes it with only 1% of data.
- PosteriorBench: tests whether generative solvers recover calibrated posteriors, not mean reconstructions, against MCMC references on physics-based inverse problems.

Dynamical systems and long-horizon consistency

- Consistency-trained operators for chaotic dynamics that preserve long-run statistics.
- PathFM: ranks first on all six trajectory-level scores and top three on fev-bench; on 135 chaotic systems, it achieves the lowest point error at every horizon and 50% lower long-run distributional error, without the variance collapse seen in baselines.

Real-world applications

- 3D urban wind prediction and energy-efficient UAV planning (NeurIPS'26 Spotlight).
- Carbon capture and storage: forward and inverse modeling with a diffusion prior.

Publications

1. **Jiachen Yao***, Zi-Siang Hsu*, Xi Deng*, Aditi Gupta, Xin Ju, Sally M. Benson, Gege Wen, and Anima Anandkumar. PosteriorBench: From Point Estimates to Posterior Matching in Evaluating Generative Inverse Solvers. *NeurIPS*, 2026. [link] [code]
2. Shaoxiang Qin, Xiongye Xiao, Yucheng Zhao, Fuyuan Lyu, Di Zhou, **Jiachen Yao**, Xue Liu, Anima Anandkumar, and Liangzhu Leon Wang. GeoWind2Plan: Mission-Time 3D Urban Wind Prediction for Energy-Efficient UAV Planning. *NeurIPS*, 2026 (**Spotlight**). [link] [code]
3. Lufang Chiang*, **Jiachen Yao***, Thomas Y.L. Lin, and Anima Anandkumar. Physics is the Best Teacher: Consistency Learning for Time-Invariant Operators of Chaotic Dynamics. *NeurIPS Workshop on AI for Stochastic Dynamics*, 2026. [link]
4. Xin Ju, **Jiachen Yao**, Anima Anandkumar, Sally M. Benson, and Gege Wen. Function-Space Decoupled Diffusion for Forward and Inverse Modeling in Carbon Capture and Storage. *ICLR Workshop on AI and PDE*, 2026. [link] [code]
5. Thomas Lin*, **Jiachen Yao***, Lufang Chiang, Julius Berner, and Anima Anandkumar. Decoupled Diffusion Sampling for Inverse Problems on Function Spaces. *ICLR Workshop on AI and PDE*, 2026 (**Oral**). [link] [code]
6. Duy Nguyen*, **Jiachen Yao***, Jiayun Wang*, Julius Berner, and Anima Anandkumar. Flow-Guided Neural Operator for Self-Supervised Learning on Time Series Data. *Transactions on Machine Learning Research*, 2026. [link]
7. Bahareh Tolooshams*, Aditi Chandrashekar*, Rayhan Zirvi*, Abbas Mammadov, **Jiachen Yao**, Chuwei Wang, Anima Anandkumar. EquiReg: Equivariance Regularized Diffusion for Inverse Problems. *Transactions on Machine Learning Research*, 2026. [link]

8. **Jiachen Yao***, Abbas Mammadov*, Julius Berner, Gavin Kerrigan, Jong Chul Ye, Kamyar Azizzadenesheli, and Anima Anandkumar. Guided Diffusion Sampling on Function Spaces with Applications to PDEs. *NeurIPS*, 2025. [link] [poster] [code]
9. Zhongkai Hao*, **Jiachen Yao***, Chang Su*, Hang Su, Ziao Wang, Fanzhi Lu, Zeyu Xia, Yichi Zhang, Songming Liu, Lu Lu, Jun Zhu. PINNacle: A Comprehensive Benchmark of Physics-Informed Neural Networks for Solving PDEs. *NeurIPS*, 2024. [link] [code]
10. **Jiachen Yao***, Chang Su*, Zhongkai Hao, Songming Liu, Hang Su, and Jun Zhu. MultiAdam: Parameter-wise Scale-invariant Optimizer for Multiscale Training of Physics-informed Neural Networks. *ICML*, 2023. [link] [poster] [code]
11. Yue Qin, Chun Yu, Wentao Yao, **Jiachen Yao**, Chen Liang, Yueting Weng, Yukang Yan, and Yuanchun Shi. Selecting Real-World Objects via User-Perspective Phone Occlusion. *CHI*, 2023. [link] [video]

Preprints:

1. **Jiachen Yao**, Xiyuan Zhang, Annan Yu, Boran Han, Abdul Fatir Ansari, Shuai Zhang, Michael Bohlke-Schneider, Bernie Wang. PathFM: Fast Generative Forecasting of Multivariate Trajectories. Under review, 2026. [link]

$*$ denotes equal contributions.

Internships

Applied Science Intern | Amazon Web Services — Jun 2026 - Sept 2026

- Developed PathFM, the first generative sampling method to rank in the top three on fev-bench, an established time series forecasting benchmark.
- Introduced path-based evaluation revealing trajectory dependencies missed by the marginal quantile forecasts used by current leading models.
- Matched the speed of leading quantile models while generating a 32-path forecast ensemble via one-step distillation and cache-friendly backbone design.

Awards and Honors

- Naren and Vinita Gupta Sensing to Intelligence Fellowship. 2026
- Professor Anatol Roshko Fellowship. 2025
- **Outstanding Graduates**, Beijing Municipal Commission of Education. 2024
- Selected to Spark Program at Tsinghua University, top 1%. 2023
- **National Scholarship**, Ministry of Education, PRC. 2022
- **Silver Medal**, National Olympiad in Informatics. 2019

Skills

Programming: PyTorch, JAX, DDP; Python, C/C++, JavaScript/TypeScript, MATLAB.

Tools: Git, Docker, Linux, CI/CD, Cloud & local deployment; FEniCS, COMSOL.

Academic Services

Teaching: Guest Lecturer, *Physics-Informed Machine Learning* (CEE 12-787), CMU, 2026; Teaching Assistant & Lecturer, *Foundations of Machine Learning and Statistical Inference* (CS 165), Caltech, 2026.

Reviewers for: *NeurIPS, ICML, ICLR, JMLR, TMLR, Nature Communications*.

Invited talks: 2nd Workshop on Learning Dynamical System (WLDS2), Japan, 2026.
Keynote for ICLR 2026 Workshop on AI & PDE, Brazil, 2026.
SIAM Conference on Uncertainty Quantification, Minneapolis, 2026.
AI & Scientific Discovery Workshop, UChicago, 2025.
Academia-Industry X Workshop, Caltech, 2024.